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H2

Started on	Tuesday, 25 August 2026, 5:07 PM
State	Finished
Completed on	Tuesday, 25 August 2026, 5:08 PM
Time taken	30 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array `nums` of size `n`, return *the majority element*.

The majority element is the element that appears more than $\lfloor n / 2 \rfloor$ times. You may assume that the majority element always exists in the array.

Example 1:

Input: `nums = [3,2,3]`

Output: 3

Example 2:

Input: `nums = [2,2,1,1,1,2,2]`

Output: 2

Constraints:

- `n == nums.length`
- `1 <= n <= 5 * 104`
- `-231 <= nums[i] <= 231 - 1`

For example:

Input	Result
3 3 2 3	3
7 2 2 1 1 1 2 2	2

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int findMajorityElement(int nums[], int n) {
4      int candidate = 0;
5      int count = 0;
6
7      // Boyer-Moore Voting Algorithm
8      for (int i = 0; i < n; i++) {
9          if (count == 0) {
10             candidate = nums[i];
11             count = 1;
12         } else if (nums[i] == candidate) {
13             count++;
14         } else {
15             count--;
16         }
17     }
18     return candidate;
19 }
20
21
22 int main() {
23     int n;
24     if (scanf("%d", &n) != 1) return 0;
25
26     int nums[n];
27     for (int i = 0; i < n; i++) {
28         scanf("%d", &nums[i]);
29     }
30
31     printf("%d\n", findMajorityElement(nums, n));
32     return 0;
33 }
```

	Input	Expected	Got	
✓	3 3 2 3	3	3	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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